

STUDENT ACTIVITY FILE

INNOVATIVE DESIGN LAB - I

Academic year: 2026-2027

Department: Computer Science and Engineering

Section:

Innovation Ecosystem Project ID: SAIRAM-IDL1-2026-CSE-09

Title of the Project: CYBERFIDEL: Autonomous Hybrid-AI SOC Analyst & Investigative Engine

Inter disciplinary Project : ☒ Yes ☐ No **Team**

Members (Max. 3)

S.No	Name	Student ID
1	Gokula Kannan M	SEC25CS196
2	Lakshan M	SEC25CS036
3	Sivabalan T	SEC25CS101

Project Version Is this project:

☐ Version 1 – New Project ☐ Version 2–Pass out Student Project(Carried Forward)

(If Version 2, provide the following details of the original project):

Previous Project Title	Sairam Innovation Project ID

Domain Name: Cybersecurity & Artificial Intelligence

IEEE Society: IEEE Computer Society

IEEE Community: IEEE Student Branch Sairam

Club and Cells: Sairam Innovation Ecosystem / Cyber Security Club

Name of the department IDL - I Coordinator: Dr. A. SHEELA

Faculty ID: IDL-FAC-01

Name of the Supervisor: Dr. A. SHEELA

Faculty ID: IDL-FAC-02

Name of the External Guide: Cyber Forensic Specialists & Industry Leads

Designation & Organization: Cyberdome Investigations / Zoho Corporation

SDG	Goal Number With Name	SAP Code with Explanation
Primary SDG:	SDG 9: Industry, Innovation, and Infrastructure	1. 1. SAP-SDG9-CYBER-01: Resilient cyber infrastructure & AI SOC automation. 2. 2. SAP-SDG9-CYBER-02: Zero-trust cloud-edge security architecture. 3.
Secondary SDG:	SDG 16: Peace, Justice, and Strong Institutions	1. 1. SAP-SDG16-SEC-01: Evidence privacy & chain-of-custody enforcement. 2. 2. SAP-SDG16-SEC-02: Citizen PII protection in digital investigation. 3.
Tertiary SDG:	SDG 8: Decent Work and Economic Growth	1. 1. SAP-SDG8-AUTO-01: Reduction of SOC analyst alert fatigue. 2. 2. SAP-SDG8-AUTO-02: Decreasing MTTR from 45 mins to < 30 seconds. 3.

S.No.	Title	Start Date	Completion Date	Mark (Out of 10)	Remarks	Signature of the IDL - I Coordinator
1	Orientation & Entrepreneurial Mindset	01/08/2026	07/08/2026		Completed	
2	Problem Identification & Passion CV	08/08/2026	14/08/2026		Completed	
3	Customer Segmentation & Persona Creation & JTBD	15/08/2026	21/08/2026		In Progress	
4	Ideation, Market Size & Competitor Analysis	22/08/2026	28/08/2026		Upcoming	
5	Milestone1:Problem-Solution Fit Presentation	29/08/2026	04/09/2026		Upcoming	
6	MVP Design & Prototyping					
7	Opportunity Assessment & Feasibility					
8	Business Model & Lean Canvas					
9	Literature survey					



10	Milestone 2 : Business Model Presentation					
11	Marketing & Sales Strategy (GTM)					
12	Financial Planning & Team Building					
13	Milestone 3 : Final Pitch- Venture Feasibility					

Outside World Projection

Details of the Competition Attended (Hackathon / Idea Pitching)

S.No	Name	Date	Place	Remark	Signature
1	Hac'KP 2026 (7th Edition National Hackathon by Kerala Police Cyberdome)	14/08/2026	Zoho Corporation, Chennai	Individual Hackathon	Finalist (Sivabalan T)
2					

Design Patent / Idea Patent Publication / Conference Paper

1	Confidentiality-Preserving Autonomous SOC Triage via Tokenized Edge-Cloud Multi-Agent Systems	August 2026	IEEE / USENIX Track	Paper Manuscript & Patent Drafted	
2					

IDL – I Coordinator
Dr. A. SHEELA

Creative Innovator
Gokula Kannan M

H.O.D
Head of Department



புயன்றால் புதுபாத்து
எதுவும் இல்லை.

என்.யு.எஸ்.
வி.பு.சு.

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PHOTOS / LITERATURE PAPERS / PATENT SEARCH / SKETCHES (PASTE OR ATTACH SEPARATE PAGES)

WEEK – 1

Security Operations Centers (SOCs) generate over 10,000 security logs daily, creating severe alert fatigue where human analysts spend 30?45 minutes manually triaging single events. Current AI supports cannot be deployed widely due to data privacy regulations and risks of uploading confidential network payloads to external servers.

1. Orientation & Entrepreneurial Mindset

The proposed solution, SENTINEL, introduces Reversible Tokenized Pseudonymization, scrubbing 100% of sensitive PII, IP addresses, and hostnames locally before AI reasoning occurs. This enables enterprise-grade autonomous triage at \$0 operational cost for 90% of routine alerts using local consumer GPU hardware (NVIDIA RTX 3050).

• Understand entrepreneurship & innovation

The global SIEM automation and threat intelligence market is valued at \$13.2 Billion+ with an 18.5% CAGR. Primary target markets include Enterprise SOCs, Managed Security Service Providers (MSSPs), Air-Gapped Defense Networks, Cyber Crime Units, and Startups.

- Technical skillset is complemented by providing 700+ parameters and down to 4-bit (168 MB) CBF format to execute high-capacity intelligence on edge workstations.

- Startup journey from idea to venture.
 1. Problem Formulation: Identifying SOC alert fatigue and data privacy constraints.
 2. Proof-of-Concept: Engineering local Zero-Trust Data Sanitizer (src/sanitizer.py).
 3. MVP Architecture: Building 3-Tier Hybrid AI Router (src/router.py).
 4. Validation & Pitching: Selected for National Pitching Round at Hac'KP 2026 @ Zoho Corporation.
 5. Commercialization: Deploying open-source SaaS framework and enterprise MSSP engine.

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WEEK – 2

Modern digital investigations face a severe bottleneck where human triage speed cannot match machine-speed cyber attacks. Public commercial cloud LLMs violate legal data privacy and chain-of-custody requirements when processing sensitive evidence.

2. Problem Identification & Passion CV

- Identify real-world problems based on personal interests

Small businesses, hospitals, and educational institutions are unable to afford \$100k+/year enterprise SIEM subscriptions (e.g., Splunk or Cortex XSOAR), leaving critical infrastructure vulnerable to ransomware. SENTINEL addresses this societal gap by delivering a self-hosted,

- Skills, experiences, and societal needs

- Prepare a Passion CV to identify potential entrepreneurial opportunities.

PASSION CV & TEAM PROFILE MATRIX:

? Gokula Kannan M (SEC25CS196 | CSE-A Core):

Domain: Software Architecture, Systems Engineering, Project Governance & Venture Strategy.

Contribution: Overall system architecture, sprint management, and project coordination.

? Lakshan M (SEC25CS036 | CSE-A Core):

Domain: Full-Stack Web Development, React.js UI, WebSockets Integrations.

Contribution: Real-time incident dashboard engineering and Human-in-the-Loop (HITL) action approval interface.

? Sivabalan T (SEC25CS101 | CSE-C Core):

Domain: AI Engine Architecture, Zero-Trust Privacy Engineering, Quantization & Vector RAG.

Contribution: Development of Zero-Trust Data Sanitizer (src/sanitizer.py) and 3-Tier AI Router (src/router.py). Individual Hac'KP 2026 Hackathon Finalist.

? Faculty Supervisor: Dr. A. SHEELA (Associate Professor, Dept. of Computer Science & Engineering).